PRESENTATION-READY SUMMARY

PulsarNav AI Presentation Summary

Slide-adaptable project narrative and technical talking points

Project	PulsarNav AI
Repository	/Users/supryo/Desktop/pulsar
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Document Class	Presentation-Ready Summary
Primary Audience	Academic reviewers, aerospace mentors, flight dynamics developers

This document describes PulsarNav AI as a scientific software system: why each component exists, the mathematics behind it, how it is implemented, and how its outcomes are validated.

Presentation-Ready Summary

1. Title

PulsarNav AI: Autonomous Pulsar-Based Deep Space Navigation Framework.

2. Problem

Deep-space spacecraft cannot depend on GPS. Ground-based DSN tracking is powerful but scheduled, delayed, and operationally expensive. Autonomous navigation is required for future lunar, Martian, and interplanetary missions.

3. Core Idea

Millisecond pulsars behave like highly stable natural clocks. A spacecraft measures pulse arrival delays from multiple pulsars. Those delays become geometric constraints on spacecraft position.

4. Key Equation

Delta t_i = (r dot n_i) / c

This equation maps position into timing delay. The inverse problem recovers spacecraft position and clock bias from multiple pulsar measurements.

5. Data Pipeline

NANOGrav PAR/TIM files are parsed into a pulsar catalog, TOA database, timing statistics, ranked pulsar lists, unit vectors, simulation outputs, plots, and dashboard-ready exports.

6. Signal Processing

The project models photon arrivals with a non-homogeneous Poisson process, folds photons into phase histograms, and estimates delays using cross-correlation, nonlinear least squares, and maximum likelihood estimation.

7. Navigation Solvers

The system includes least squares, weighted least squares, a 10-state error-state Kalman filter, and an 8-state absolute EKF with RK4 propagation and J2 perturbation support.

8. Validation

Monte Carlo simulation measures position error across timing noise levels, pulsar counts, and mission regions. CRLB analysis provides theoretical estimator limits.

9. Dashboard

The Next.js dashboard provides simulation controls, charts, CSV/PDF style reporting, 3D space visualization, and animated trajectory playback.

10. Comparative Lab

The comparison engine evaluates DSN-only, XNAV-only, and Hybrid EKF navigation with DSN range/Doppler, pulsar measurements, and fused estimates.

11. Achievements

- GPS-independent spacecraft navigation simulation.
- Real pulsar catalog and timing-data integration.
- Mathematical implementation of pulsar signal, delay, and navigation geometry.
- Research dashboard with interactive aerospace visualization.
- Validation tools suitable for project review and future publication work.

12. Limitations

Some UI panels remain experimental, full ATNF/JPL ephemeris integration is future work, and real detector physics can be made more detailed.

13. Future Work

Add ATNF catalog ingestion, JPL DE440 ephemerides, covariance ellipsoids, real detector response models, star-tracker fusion, optical navigation, and production-grade hybrid navigation studies.

14. Closing

PulsarNav AI demonstrates the full research chain from astrophysics to flight-dynamics software: pulsar timing data, signal processing, estimator mathematics, Monte Carlo validation, and mission-control visualization.